



Yeshwantrao Chavan college of Engineering, Nagpur
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By

Sushant Ambekar, Roll No: 155

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Guide

Prof. Pratibha Waghale

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1. Problem Statement

- Modern web applications require scalable, reliable, and fast deployment pipelines along with efficient frontend and backend integration. Manual deployment methods are error-prone and time-consuming.
- During the internship, the problem addressed was to develop a full-stack MERN application and automate its deployment using DevOps practices to ensure smooth integration, testing, and delivery.

2. Objectives & Scope

- To learn core web technologies to build responsive and user-friendly interfaces using HTML, CSS, and JavaScript.
- To develop single-page applications using React with components, forms, routing, and state management.
- To build backend services using Node.js and Express, and perform CRUD operations using MongoDB.
- To automate deployment using Git, Docker, and CI/CD pipelines to deploy MERN applications efficiently.

3. Methodology

Followed Agile software development methodology.

- Requirement analysis and task breakdown.
- Full-stack application development in iterations.
- API integration between frontend and backend
- CI/CD pipeline creation for automated deployment
- Testing and monitoring of deployed applications

4. Tools/Technologies Used

- Development
 - React.js
 - Node.js
 - Express.js
 - MongoDB
- DevOps & Deployment
 - Git & GitHub
 - Jenkins / GitHub Actions (CI/CD)
 - Docker (containerization)
 - Cloud platforms (AWS / Azure)
- Other Tools
 - Visual Studio Code
 - Postman
 - npm

5. Implementation & Outcome

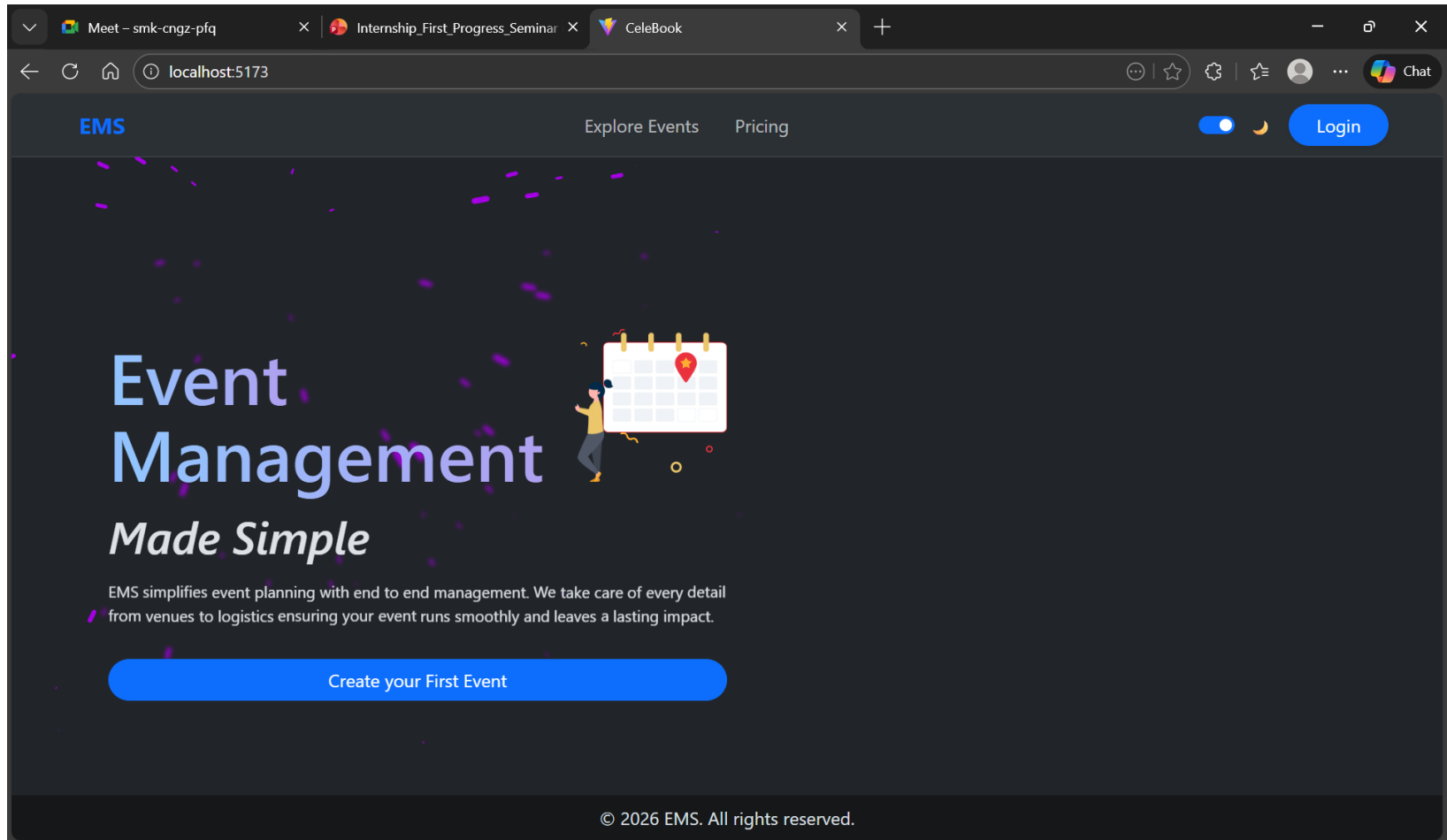
- **Implementation**

- Currently undergoing training to design and develop responsive web user interfaces using HTML5, CSS3, and JavaScript, and building single-page applications using Angular and React.
- Backend development using Node.js and Express is being implemented with MongoDB for CRUD operations.
- DevOps practices such as Git-based version control, Docker containerization, and CI/CD pipeline setup using Jenkins are also being learned and applied for application deployment.

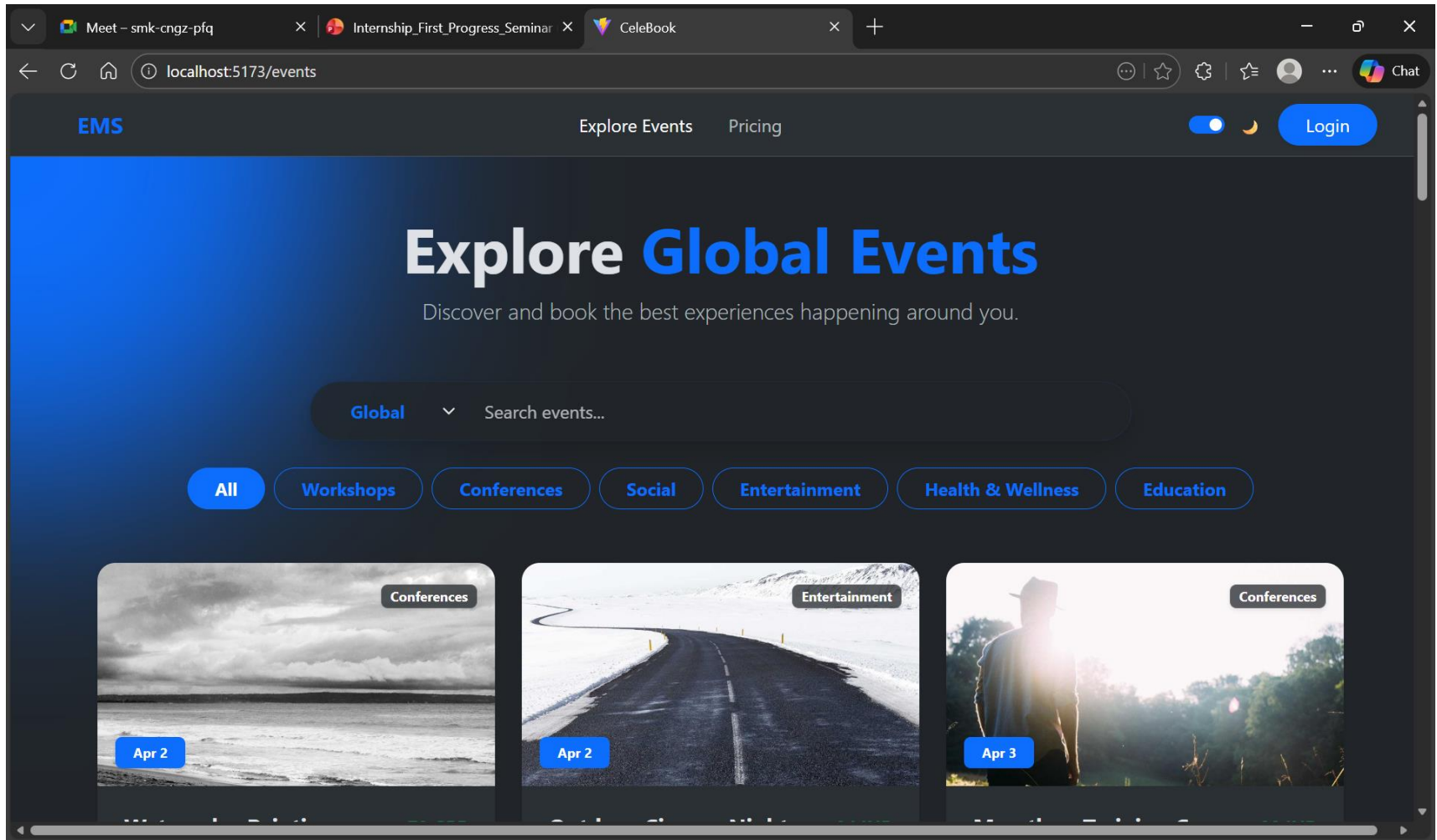
- **Outcome**

Upon completion, the training will enable the development of complete MERN stack web applications with efficient frontend–backend integration, automated deployment pipelines, early error detection, and scalable deployment following industry-standard DevOps practices.

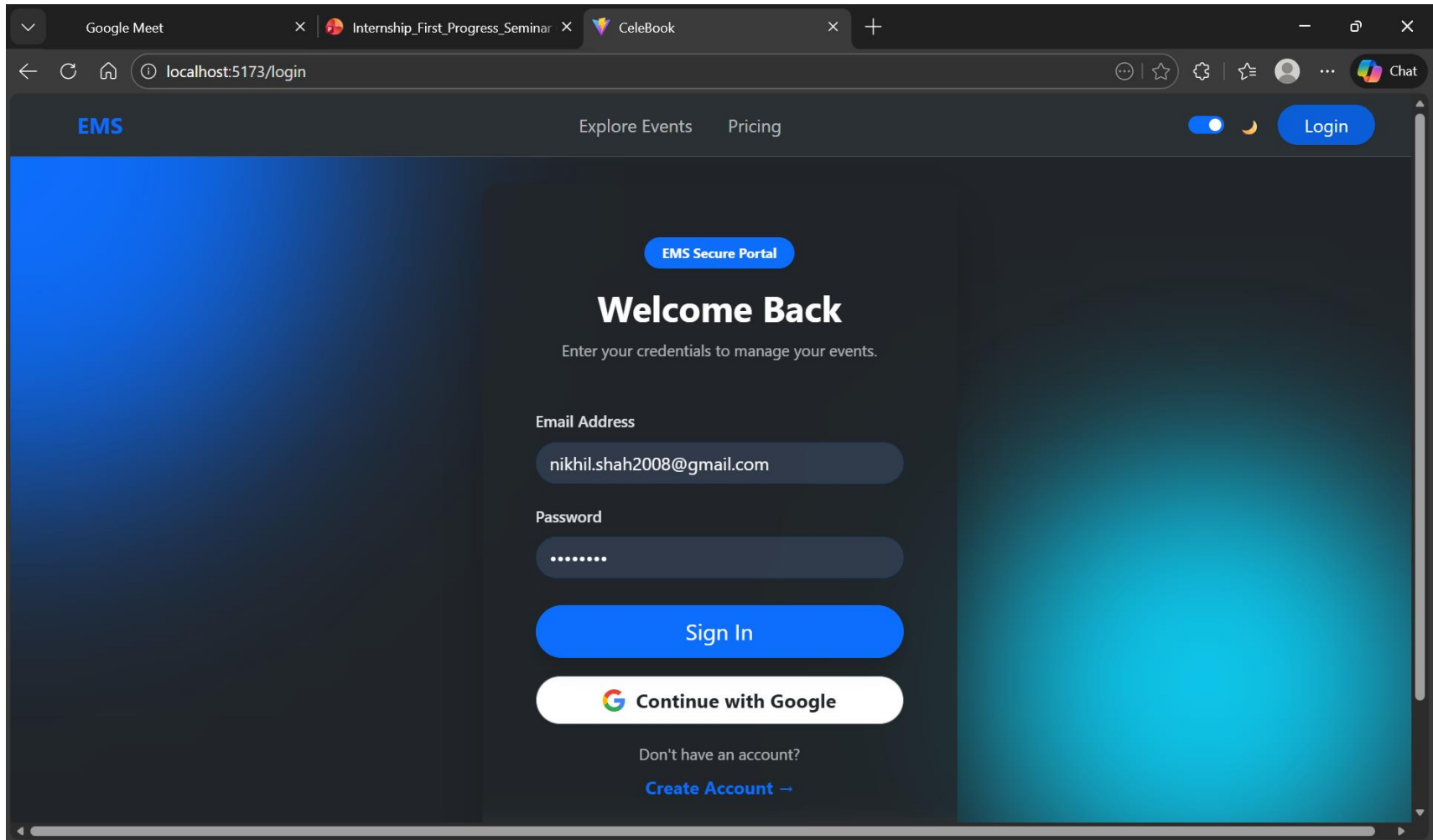
5. Implementation & Outcome



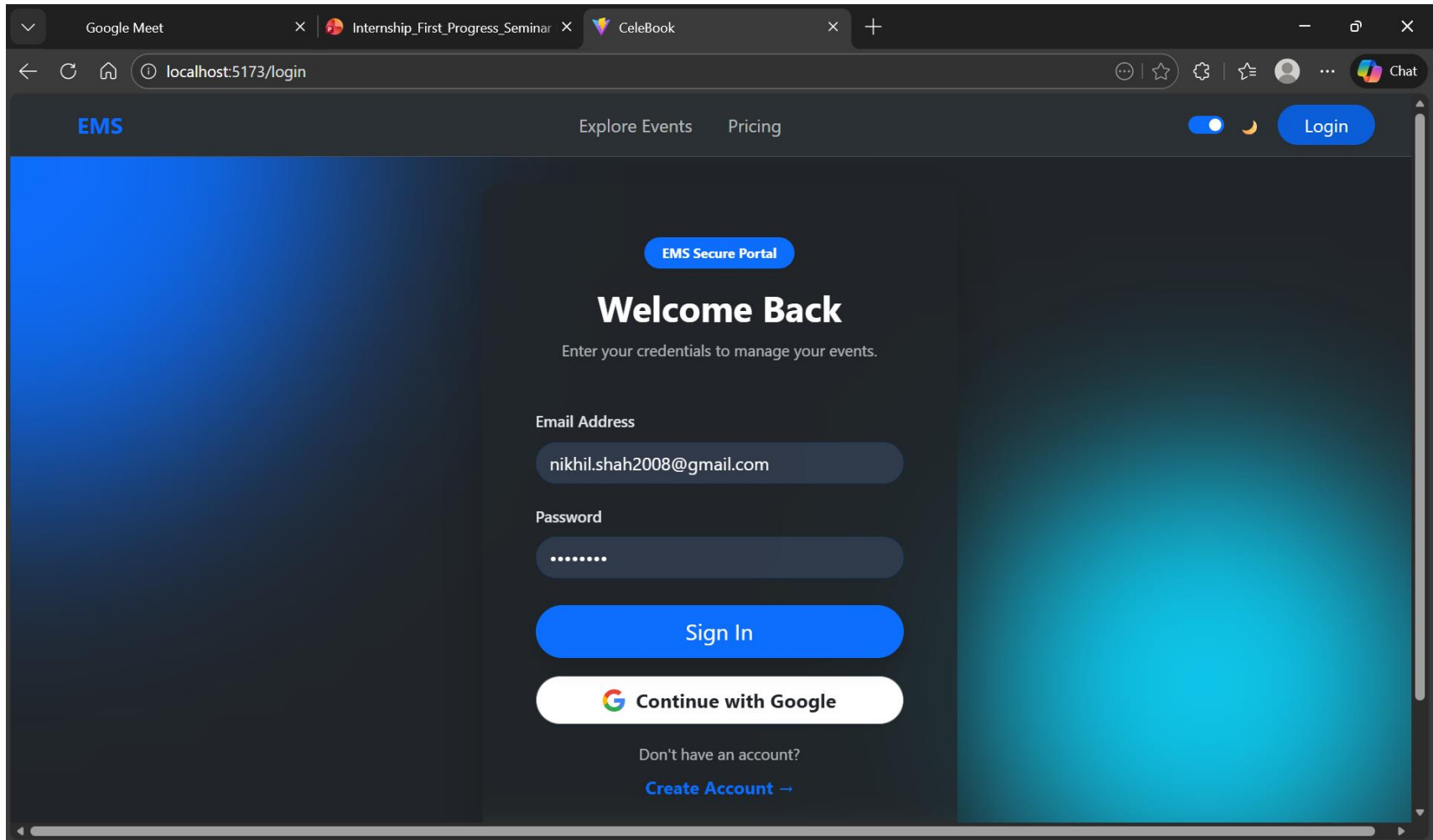
5. Implementation & Outcome



5. Implementation & Outcome



5. Implementation & Outcome



6. Outcome of Internship

- Strong understanding of MERN stack development
- Gained practical knowledge of DevOps and CI/CD pipelines
- Learned containerization and cloud deployment
- Improved teamwork, debugging, and professional communication skills
- Exposure to real-world enterprise application workflows

7.References

- <https://react.dev>
- <https://nodejs.org>
- <https://expressjs.com>
- <https://www.mongodb.com>
- <https://www.docker.com>
- <https://www.udemy.com>



Thank
you

